

DMITRII NIKOLAEV

Senior AI Engineer | Generative & Agentic AI | Ph.D. in Applied Maths

✉ dmitrynm@fastmail.com | 📞 +358 41 325 8839 | 🔗 linkedin.com/in/dmitrynm | 🏠 gitlab.com/dmitrynm

PROFILE

AI engineer building generative and agentic AI systems end to end — from the application layer (RAG pipelines, multi-agent workflows, fine-tuning, rigorous evaluation) down to the serving and training infrastructure beneath it. I take LLM applications from idea through POC to deployed, monitored product, and my systems-level depth — distributed training, inference optimization, Kubernetes — is what keeps them reliable, measurable, and efficient in production.

Ph.D. in Applied Mathematics, 7+ years delivering production ML/AI systems in Python (plus occasional C++, JavaScript, Java, and Go), and 7 years in academia.

KEY SKILLS

- GenAI applications: LLM application development with hosted LLM APIs and self-hosted open-weight models, RAG pipelines (vector databases, hybrid retrieval, LLM re-ranking), LoRA fine-tuning, prompt engineering, structured evaluation pipelines for output quality.
- Agentic systems: design and evaluation of multi-agent workflows, tool-use orchestration, MCP integration, agent harness engineering; daily AI-assisted development with Claude Code / Cursor, including validation of AI-generated content.
- Software engineering: scalable, maintainable codebases — classical and modern design patterns (SOLID, GoF, FP, dependency injection, event-driven and async architectures), refactoring, code review.
- Production engineering & MLOps: Python, FastAPI / REST web services, Go and Java for data-focused microservices and APIs, React and Vue for data dashboards and analytics frontends, Docker, Kubernetes, Helm, CI/CD, unit testing, monitoring and observability, MLFlow, Azure and AWS; data pipelines and integrations across diverse data sources and formats.
- ML stack: PyTorch, HF Transformers/Diffusers, vLLM, SGLang, LightGBM, scikit-learn, NumPy, Pandas; C++ where performance demands it.
- LLM infrastructure: distributed training (DeepSpeed, Megatron-LM, multi-node GPU orchestration) and inference (KV-cache management, dynamic batching, attention optimizations).
- Model documentation and reproducibility: benchmarking, experiment tracking, evaluation reports.

SELECTED EXPERIENCE

Silo AI at AMD

Staff AI Engineer

Helsinki, Finland

Aug 2024 – Present

- Building an agentic harness for intrusive testing and knowledge organization — multi-agent workflows driven by open-weight LLMs served through self-hosted vLLM endpoints, in which agents modify code, allocate cloud resources, run experiments, and assemble reports and storyboards; tool use exposed via custom MCP servers, with systematic evaluation of agent behavior (internal AMD tooling).
- Deployed and optimized vLLM-based LLM inference in production — unblocked concurrent inference of the 8-bit-quantized MiniMax 2.5 model by resolving memory-allocation failures and tuning allocation strategies; researching interactivity of long-context multi-turn agentic pipelines (client: AWS).
- Operated the client's production distributed-training workloads on multi-GPU clusters; built monitoring and observability tooling that improved reliability of live training and inference pipelines (client: Samsung).
- Ported PyTorch model inference to AMD's next-generation FPGA accelerators and XDNA spatial AI-engine arrays, purpose-built for AI workloads — delivered as a component of the client's efficient-inference platform; adapted models and operators to hardware constraints and validated correctness and performance (client: Microsoft).

Silo AI (AI R&D Lab)

Senior AI Engineer

Nov 2023 – Aug 2024

- Delivered a predictive pricing solution for an enterprise client — gradient-boosting models (LightGBM) built and deployed on Azure AI Foundry and served through managed Azure AI endpoints, from feasibility exploration with business stakeholders to a working product (client: Assa Abloy).
- Shipped a patent knowledge-retrieval system for a client project — PostgreSQL-backed RAG with hybrid retrieval and LLM re-ranking, taken from feasibility prototype to delivered product with the business stakeholders (client: Nokia).
- Prototyped a suite of LLM-based platform components for code generation, AI-vs-human code detection, and coding-standards compliance checking — with evaluation pipelines to rigorously measure detection quality (client: SEMA Technologies).
- Developed core functionality of an in-house text-to-image platform and deployed it as a production service; applied LoRA fine-tuning and multi-LoRA layer merging to achieve target style compositions without retraining (product: SiloSnap).

Skolkovo Institute of Science and Technology (AI R&D Lab)

Moscow, Russia

Software / Research AI Engineer

Jul 2019 – Apr 2023

- Built an end-to-end RL training pipeline for multi-agent decision-making in sequential games — self-play rollout generation, opponent modeling with adaptive strategy selection, and policy evaluation against a legacy baseline — reaching parity with the expert-authored rule-based system and removing the dependency on hand-crafted rules (client: deeplay.io).
- Designed and built a multi-modal analytics platform for an international startup — fused telemetry, video, text, and biometric signals into real-time dashboards used in live events; anomaly detection over heterogeneous behavioral data (client: HeadKraken).
- Integrated tensor-storage data lakes into distributed training pipelines — tuned PyTorch data loaders and migrated datasets, raising dataset availability and removing the input bottleneck for GPU training jobs (client: activeloop.io).

FusionSoft (Software Development Studio)

Lipetsk, Russia

Software Engineer

Dec 2017 – Mar 2019

- Built and debugged GPU job orchestration for an international AI platform startup, on its Kubernetes-native MLOps platform for distributed training and synthetic-data generation (client: Neuromation.io).
- Designed and developed a scalable, high-load enterprise analytics platform replacing legacy SAP modules for an international client; reduced analytics query time from days to seconds (client: La Lumière).

Lipetsk State Technical University

Lipetsk, Russia

Assistant / Associate Professor

Feb 2010 – Dec 2017

- Taught ML, optimization, numerical methods, and probability theory for 7 years; supervised 8 graduate students and authored 7 peer-reviewed publications, presented talks at 20+ international and regional conferences.
- Built ML research infrastructure and delivered applied prototypes to transportation and logistics partners; improved robotic navigation flexibility on manufacturing lines by replacing hardcoded routes with neural vision-based localization (client: Indesit).

EDUCATION

Voronezh State Technical University | *Ph.D. in Applied Mathematics*

2010–2014

Thesis: numerical methods for modelling and control of multi-agent systems — developed neural-network-driven coordination and control of cooperating agents.

Lipetsk State Technical University | *Specialist in Applied Mathematics*

2005–2010

Five-year degree, equivalent to B.Sc. + M.Sc. Thesis: computer vision and algebraic methods for autonomous robot control — built neural-network control of a robotic hand under physics-based constraints.

ADDITIONAL INFORMATION

Location Helsinki, Finland (hybrid with office days · mobile within EU · open to remote)

Languages English (professional), Russian (native), Finnish (conversational)